

Installation and Commissioning Quality Control Record

Rotor pole concentricity, circularity and verticality, rotor diameter

 ITP step #: 6300-17
 Project: GMS rotor

 Unit : 4

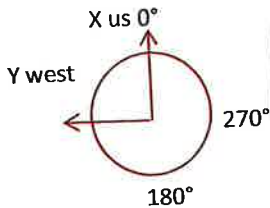
 Revision # 01
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 Number of readings: 8

 Readings in : inch

Rotor hub level

M_PL_COUPLING_FLANGE_OVER_WEB (Measured Plane , number of points 8)



Center	X	-0.527
	Y	-0.325
	Z	3.260
Normal	I	0.000020
	J	-0.000024
	K	-1.000000
RMS		0.002
Standard Dev		0.002
Max		0.002
Min		-0.003
Form		0.005

Readings

X	Y	RAD bottom	ANGLE	Z	Dev Fit
54.664	20.473	58.372	21	3.259	0.002
24.177	53.189	58.426	66	3.258	0.001
-22.338	53.239	57.736	113	3.262	-0.003
-53.352	24.434	58.681	155	3.257	0.001
-52.956	-24.781	58.467	205	3.258	0.001
-28.952	-51.009	58.653	240	3.262	-0.001
21.203	-54.429	58.413	291	3.261	0.001
53.335	-23.716	58.370	336	3.264	-0.002

	Tolerance	Results
LEVEL		0.005

Readings taken where rotor was shimmed (over stand webbing)

NCR #: _____

 Measurement Instruments
LASER TRACKER
X0100802531

 Before: 26 °C
 After: 27 °C

 Measured by: Pascal Mitre
 ANDRITZ Rep.: Mickeal Kelly
 Cust. Repres.: Kathy

 dd-mmm-yyyy
 sept 19 2012

Katie Taylor

Installation and Commissioning
Quality Control Record

**Rotor pole concentricity, circularity and verticality,
rotor diameter**

Q-6301-041

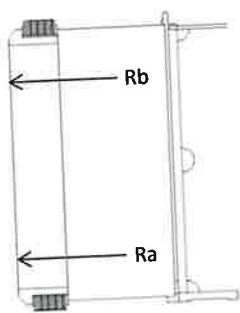
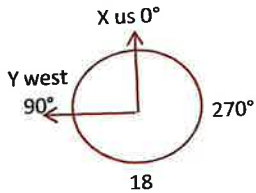
ITP step #: 6300-17
Project: GMS rotor

Unit : 4

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Number of readings: 40

Readings in : inch



Note:
Readings taken at
150mm (6in) from the top
and the bottom

M_CI_RB_BOTTOM (Measured Circle , number of points 48)

Measured			
Best Center	X	0.000	
	Y	0.000	
	Z	0.000	
Diameter		428.516	
	I	0.000000	
	J	0.000000	
Normal	K	1.000000	
	RMS	0.007	
Standard Dev		0.007	
	Max	0.012	
	Min	-0.014	tol avg
Circularity		0.026	0.046

M_CI_RA_TOP (Measured Circle , number of points 48)

Measured			
Best Center	X	0.001	
	Y	-0.001	
	Z	59.295	
Diameter		428.522	
	I	0.000000	
	J	0.000000	
Normal	K	1.000000	
	RMS	0.007	
Standard Dev		0.007	
	Max	0.014	
	Min	-0.011	tol avg
Circularity		0.025	0.046

On poles with airgap monitors, readings were taken below airgap

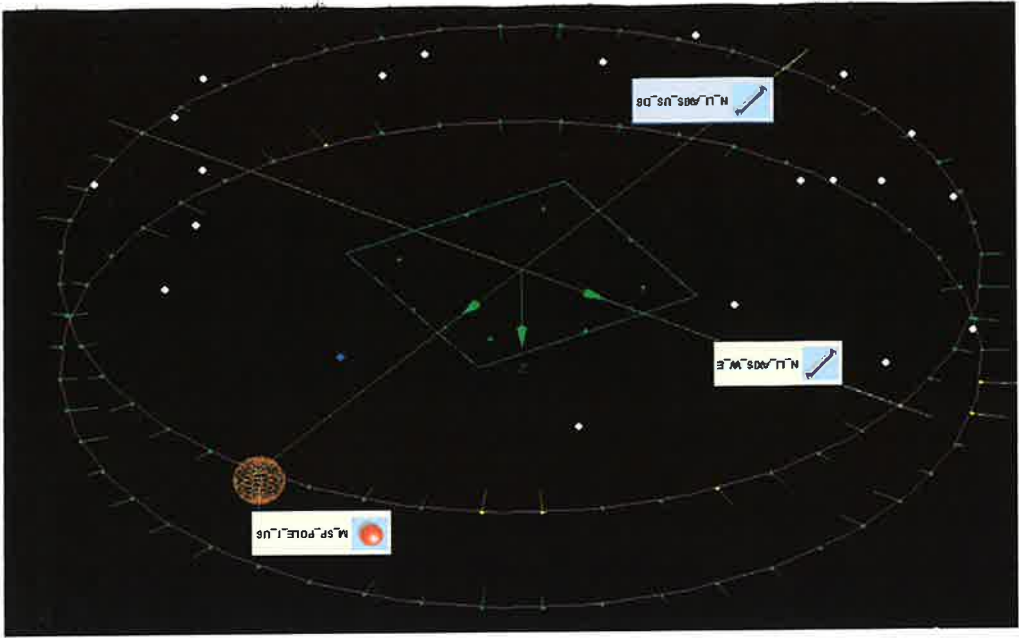
NCR #: _____

Measurement Instruments	T°	dd-mmm-yyyy
LASER TRACKER	Before: 26 °C	sept 19 2012
X0100802531	After: 27 °C	
	Measured by: Pascal Mitre	
	ANDRITZ Rep.: Mickeal Kelly	
	Cust. Repres.: Kathy	

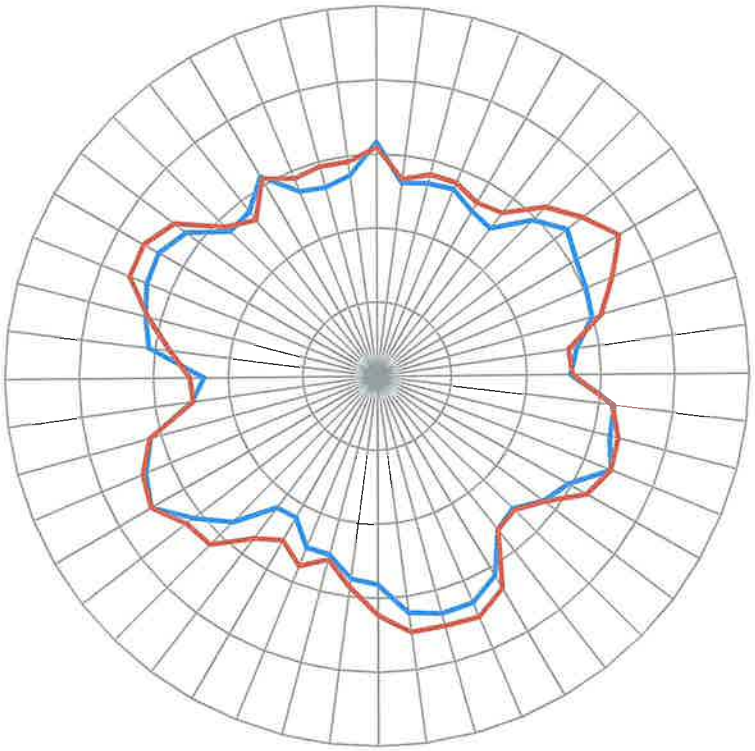
M_CI_RB_BOTTOM (Measured Circle , number of points 48)					
	X	Y	RAD bottom	ANGLE	Z
1	214.256	0.323	214.256	0	0.000
2	212.459	27.683	214.255	7	0.000
3	207.036	55.128	214.250	15	0.000
4	198.059	81.705	214.250	22	0.000
5	185.550	107.105	214.244	30	0.000
6	170.222	130.097	214.245	37	0.000
7	151.425	151.577	214.255	45	0.000
8	130.355	170.047	214.262	53	0.000
9	106.956	185.666	214.270	60	0.000
10	82.021	197.947	214.267	67	0.000
11	55.603	206.923	214.263	75	0.000
12	28.176	212.389	214.250	82	0.000
13	-0.029	214.247	214.247	90	0.000
14	-27.643	212.471	214.262	97	0.000
15	-55.331	206.997	214.264	105	0.000
16	-82.038	197.939	214.267	113	0.000
17	-107.316	185.456	214.267	120	0.000
18	-130.567	169.887	214.264	128	0.000
19	-151.723	151.280	214.256	135	0.000
20	-170.110	130.262	214.256	143	0.000
21	-185.626	107.011	214.262	150	0.000
22	-197.814	82.307	214.254	157	0.000
23	-206.983	55.337	214.253	165	0.000
24	-212.360	28.432	214.255	172	0.000
25	-214.263	0.505	214.263	180	0.000
26	-212.398	-28.127	214.253	188	0.000
27	-206.958	-55.434	214.254	195	0.000
28	-197.968	-81.936	214.254	202	0.000
29	-185.483	-107.236	214.251	210	0.000
30	-169.959	-130.449	214.250	218	0.000
31	-151.517	-151.491	214.259	225	0.000
32	-130.418	-170.001	214.264	233	0.000
33	-107.224	-185.502	214.262	240	0.000
34	-82.218	-197.858	214.261	247	0.000
35	-55.467	-206.956	214.260	255	0.000
36	-28.127	-212.400	214.254	262	0.000
37	-0.023	-214.252	214.252	270	0.000
38	27.640	-212.474	214.264	277	0.000
39	55.432	-206.970	214.264	285	0.000
40	81.937	-197.982	214.268	292	0.000
41	106.9338	-185.667	214.259	300	0.000
42	130.380	-170.020	214.256	307	0.000
43	151.490	-151.506	214.251	315	0.000
44	169.946	-130.471	214.253	322	0.000
45	185.535	-107.168	214.263	330	0.000
46	197.922	-82.079	214.266	337	0.000
47	206.962	-55.471	214.266	345	0.000
48	212.405	-28.165	214.264	352	0.000

M_CI_RA_TOP (Measured Circle , number of points 48)					
X	Y	RAD top	ANGLE	Z	top - bottom rad
214.265	0.180	214.265	0	59.295	0.008
212.463	27.675	214.258	7	59.295	0.002
207.039	55.122	214.251	15	59.295	0.001
198.120	81.570	214.255	22	59.295	0.005
185.705	106.851	214.251	30	59.295	0.007
170.122	130.244	214.255	37	59.295	0.010
151.706	151.308	214.264	45	59.295	0.009
130.294	170.096	214.265	53	59.295	0.002
107.054	185.610	214.270	60	59.295	0.000
81.871	198.010	214.268	68	59.295	0.001
55.316	207.000	214.263	75	59.295	0.000
27.900	212.425	214.250	83	59.295	0.000
0.175	214.250	214.250	90	59.295	0.004
-28.006	212.418	214.256044	98	59.295	-0.006
-55.634	206.915	214.263567	105	59.295	-0.001
-82.066	197.933	214.271746	113	59.295	0.005
-106.883	185.711	214.272	120	59.295	0.005
-130.243	170.140	214.268	127	59.295	0.004
-151.343	151.663	214.257	135	59.295	0.002
-170.286	130.028	214.253	143	59.295	-0.003
-185.657	106.956	214.262	150	59.295	-0.001
-198.187	81.415	214.258	158	59.295	0.004
-207.001	55.295	214.259	165	59.295	0.006
-212.481	27.540	214.258	173	59.295	0.004
-214.262	-0.112	214.262	180	59.295	-0.001
-212.458	-27.677	214.254	187	59.295	0.001
-207.038	-55.145	214.256	195	59.295	0.003
-197.907	-82.088	214.256	203	59.295	0.002
-185.466	-107.270	214.254	210	59.295	0.003
-169.850	-130.599	214.255	218	59.295	0.005
-151.379	-151.636	214.264	225	59.295	0.005
-130.345	-170.064	214.270	233	59.295	0.005
-107.037	-185.625	214.275	240	59.295	0.013
-82.080	-197.924	214.268	247	59.295	0.008
-55.508	-206.947	214.262	255	59.295	0.003
-28.114	-212.399	214.252	262	59.295	-0.002
-0.243	-214.252	214.252	270	59.295	0.001
27.847	-212.446	214.263	277	59.295	-0.001
55.404	-206.980	214.267	285	59.295	0.002
81.851	-198.018	214.268	292	59.295	0.000
106.800	-185.750	214.265	300	59.295	0.006
130.539	-169.898	214.256	308	59.295	0.000
151.339	-151.658	214.252	315	59.295	0.001
169.970	-130.440	214.253	322	59.295	0.000
185.514	-107.213	214.267	330	59.295	0.004
197.909	-82.123	214.271	337	59.295	0.004
206.959	-55.493	214.270	345	59.295	0.003
212.446	-27.895	214.270	353	59.295	0.005

max 0.013
min -0.006
average 0.003



		Vertically (max)
	0.046	Circularity (average top and bottom)
	0.061	Circularity (individual)
	0.009	Concentricity (average)
Result	0.013	
	0.026	
	0.031	
	0.001	



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Installation and Commissioning
Quality Control Record**Rotor Poles Hipot & Megger Test****Q-0500-002**ITP step #: 6300-18Revision # 00Project: GMS Generating StationUnit : 4Page 1 of 2**SAFETY CHECK LIST**

Equipment Tested: Gavin Frie, Geordie Barr Test voltage: 5000 ☐ Volt AC
Test Leader: Gavin Frie, Geordie Barr ☒ Volt DC
Equipment operator: Gavin Frie, Geordie Barr
Power ON checked by: Gavin Frie 7,500 Volt DC on Hipot
Power OFF checked by: Gavin Frie
Watchman: George McIver Location: Rotor Poles
First Aid Coverage by: Geordie Barr

Components to be grounded: Rotor Spider
Pole Instrumentation cables
Distance of approach: 3 Ft
Theoretical grounding duration time: 1 minute

Grounding done: ☒ Yes ☐ No ☐ N/A
Insulated rubber gloves available: ☒ Yes ☐ No ☐ N/A
Barrier tape in place: ☒ Yes ☐ No ☐ N/A
LOTO Locks available: ☐ Yes ☐ No ☒ N/A
Ground stick is checked and ready: ☒ Yes ☐ No ☐ N/A
Required PPE is available and used: ☒ Yes ☐ No ☐ N/A

Additional checks and comments:

Individual testing of each pole installed on rotor prior to horseshoe connection installation5000 V DC on megger7,500 V DC on hipotFlashing lights and signage used to provide testing in progress awareness.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Gavin Frie</u>	dd-mmm-yyyy
Megger BM11 #16008	Before: <u>27 C</u>	<u>Geordie Barr</u>	<u>4-Sep-2012</u>
Hipot Tester #416362	After: <u>27 C</u>	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>4-Sep-2012</u>
		Cust. Repres. <u>[Signature]</u>	<u>Sept 27, 12</u>

Installation and Commissioning Quality Control Record

Rotor Poles Hipot & Megger Test

Q-0500-002

 ITP step #: 6300-18

 Revision # 00

 Project: GMS Generating Station

 Unit : 4

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 Number of readings: 48

 Readings in : G Ohms

 Humidity recorded at 43%

Pole	Megger test	Hipot test	Pole	Megger test	Hipot test
1	16.4	Passed	25	16.4	Passed
2	16.4	Passed	26	20.5	Passed
3	20.5	Passed	27	16.4	Passed
4	30.8	Passed	28	28.7	Passed
5	16.4	Passed	29	41.0	Passed
6	16.4	Passed	30	41.0	Passed
7	14.4	Passed	31	16.4	Passed
8	16.4	Passed	32	41.0	Passed
9	20.5	Passed	33	41.0	Passed
10	20.5	Passed	34	41.0	Passed
11	20.5	Passed	35	41.0	Passed
12	20.5	Passed	36	14.4	Passed
13	20.5	Passed	37	20.5	Passed
14	20.5	Passed	38	41.0	Passed
15	41.0	Passed	39	41.0	Passed
16	20.5	Passed	40	41.0	Passed
17	8.2	Passed	41	41.0	Passed
18	41.0	Passed	42	41.0	Passed
19	41.0	Passed	43	41.0	Passed
20	41.0	Passed	44	14.4	Passed
21	41.0	Passed	45	41.0	Passed
22	41.0	Passed	46	8.2	Passed
23	41.0	Passed	47	20.5	Passed
24	41.0	Passed	48	16.4	Passed

Values Corrected at 40 C. Pole # 28, 36, 44 were re-tested after applying heat to remove humidity

Notes: 1- Megger test at

5000 V DC

 Acceptance criteria is higher than 1 Gigaohms.

2- Hipot test at

7500 V DC

for 1 minute. all connected.

NCR #: _____

Pole #28,36,44 have low values at 5000V however they average 50 G ohms at 1000V

Measurement Instruments	T°	Measured by:	Gavin Frie	dd-mmm-yyyy
Megger BM11 #16008	Before: <u>27 C</u>		Geordie Barr	<u>27-Aug-2012</u>
Hipot Tester #416362	After: <u>27 C</u>	ANDRITZ Rep.:	Jean-Luc Cote	<u>27-Aug-2012</u>
		Cust. Repres.	<u>[Signature]</u>	<u>Sept 27, 12</u>

Installation and Commissioning

Quality Control Record

Torque Monitoring

Q-0000-003B

ITP step #: 6300-19,21,22

Revision # 00

Project: GM Shrum

Unit : 4

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[illegible]

Note: Rotor pole field leads are now installed on rotor rim. Remainder of field leads will only be installed after the rotor and collector rings are installed. All installed hardware was locked with locking tabs

NCR #: _____

Measuring Instrument	T°	Measured by:	George McIver	dd-mmm-yyyy
Torque Wrench #419008	Before: 25.4 C			
Torque Wrench #418006	After: 25.4 C	ANDRITZ Rep.:	Mike Kelly	18-Sep-2012
		Cust. Repres.	<i>[Signature]</i>	<i>[Signature]</i>

Installation and Commissioning

Quality Control Record

Horseshoe connection radial gap

Q-6323-010

 ITP step #: 6300-20

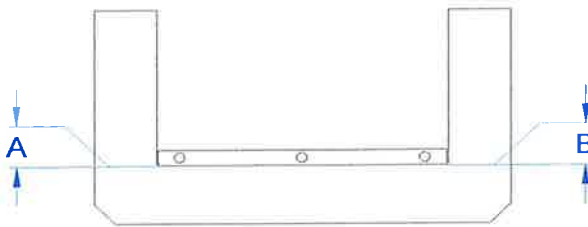
 Revision # 01

 Project: GM Shrum

 Unit : 4

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 Number of readings: 24

 Readings in : mm


NOTE: Take gap with filler gauge on 50mm (2") from each corner.
 Gap should be zero in these areas.

Conn #	Radial Gap			
	Top Horseshoe		Bottom Horseshoe	
	A	B	A	B
1	0.00	0.00	0.00	0.00
2	0.00	0.00	0.00	0.00
3	0.00	0.00	0.00	0.00
4	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00
6	0.00	0.00	0.00	0.00
7	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00
10	0.00	0.00	0.00	0.00
11	0.00	0.00	0.00	0.00
12	0.00	0.00	0.00	0.00
13	0.00	0.00	0.00	0.00
14	0.00	0.00	0.00	0.00
15	0.00	0.00	0.00	0.00
16	0.00	0.00	0.00	0.00
17	0.00	0.00	0.00	0.00
18	0.00	0.00	0.00	0.00
19	0.00	0.00	0.00	0.00
20	0.00	0.00	0.00	0.00
21	0.00	0.00	0.00	0.00
22	0.00	0.00	0.00	0.00
23	0.00	0.00	0.00	0.00
24	0.00	0.00	0.00	0.00

NCR #: _____

Measurement Instruments	T°	Measured by: <u>George Mciver</u>	dd-mmm-yyyy
<u>Feeler Gauge</u>	Before: _____		<u>19-Sep-2012</u>
	After: _____	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	
		Cust. Repres. _____	

Installation and Commissioning
Quality Control Record**Megger, PI, and Hipot Test****Q-0500-001**ITP step #: 6300-23

Revision # 00

Project: GM ShrumUnit : 4

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SAFETY CHECK LIST

Equipment Tested: Yes Test voltage: 7500 ☐ Volt AC
Test Leader: Gavin Frye ☒ Volt DC
Equipment operator: Gavin Frye
Power ON checked by: Gavin Frye
Power OFF checked by: Gavin Frye
Watchman: George Mciver Location: Rotor on floor of powerhouse
First Aid Coverage by: Geordie Barr

Components to be grounded: Rotor field polesDistance of approach: 1 meterTheoretical grounding duration time: 8hrs

Grounding done: ☒ Yes ☐ No ☐ N/A
Insulated rubber gloves available: ☒ Yes ☐ No ☐ N/A
Barrier tape in place: ☒ Yes ☐ No ☐ N/A
LOTO Locks available: ☒ Yes ☐ No ☐ N/A
Ground stick is checked and ready: ☒ Yes ☐ No ☐ N/A
Required PPE is available and used: ☒ Yes ☐ No ☐ N/A

Additional checks and comments:

Rotor rim grounded to station grounding bus bar.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Gavin Frye</u>	dd-mmm-yyyy
Dielectric test set	Before: <u>26°</u>		<u>24-Sep-2012</u>
model HVD 120/5	After: <u>26°</u>	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	
#416362		Cust. Repres. _____	

Installation and Commissioning Quality Control Record

Megger, PI, and Hipot Test

Q-0500-001

ITP step #: 6300-23

Revision # 00

Project: GM Shrum

Unit : 4

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Inspection

Component: Rotor field poles

Humidity level %: 45

☒ An examination for cleanliness was performed and found to be satisfactory.

Authorized for acceptance test:

Mike Kelly

24-09-12

Andritz technical advisor

Date

☐ **Polarisation index:** Polarisation index = (Megohms at 10 min)/(Megohms at 1 min)

DC voltage: _____

Units: _____

Time	Readings
0 min	
1 min	
10 min	
10min/1min	

Note: If after 1 minute, resistance is higher than 5 giga-ohms, PI is irrelevant (IEEE 43-2000).

Minimum

2

☒ Megger test

Criterion for acceptance: 200MΩ

Voltage : 2500

Humidity % : 45

Resistance to ground : 525MΩ

Resistance corrected at (°C): 215MΩ @ 40 C

Test duration : 1 minute

Acceptable ☒

Non-conforming ☐

☒ Hipot test

Voltage : 7500

Humidity % : 45%

Leakage current (Hipot DC) : 36 μ amps

Test duration : 1 Minute

Acceptable ☒

Non-conforming ☐

Test performed by : Gavin Frye

24-09-12

Date

NCR #: _____

Measurement Instruments	T°	Measured by:	Gavin Frye	dd-mmm-yyyy
Dielectric test set	Before: 27°		George Mciver	24-Sep-2012
model HVD 120/5	After: 27°	ANDRITZ Rep.:	Jean-Luc Cote	25-Sep-2012
#416362		Cust. Repres.		

Installation and Commissioning Quality Control Record

Rotor impedance (pole drop) test

Q-GMS-6325-030

ITP step #: 6300-24

Revision # 00

Project: GM Shrum

Unit : 4

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Number of readings: 48

Readings in : Volts

[illegible]

Test frequency: 60

Voltage applied 96Kvac

NCR #:

Variation to average	
Average	1.997708333
Tolerance (%)	5%
Maximum variation with average	0.98

*Variation to average= ((Volt-Aver.)/Aver.)x100

Measurement Instruments	T°	Measured by:	Gavin Frye	dd-mmm-yyyy
Multimeter Fluke 77	Before: 27°c		George Mciver	23-Sep-2012
	After: 27°c	ANDRITZ Rep.:	Mike Kelly	24-Sep-2012
humidity	45%	Cust. Repres.		

Installation and Commissioning Quality Control Record

ITP closure

Q-0000-002

ITP step #: 6300-26

Revision # 00

Project: GMS

Unit : 4

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THIS FORM MUST BE COMPLETED BY THE QA ADVISOR OR HIS REPRESENTATIVE.

THIS FORM MUST BE COMPLETED AT THE END OF THE WORK FOR EACH SECTION OF THE ITP.

ALL NON-CONFORMANCES, DEFICIENCIES, ETC. MUST BE CORRECTED PRIOR TO THE PREPARATION OF THIS REPORT.

THIS INSPECTION COVERS ONLY THE ITEMS SPECIFIED IN THE CONTRACT.

	NO	YES	N/A
ALL WORK IS COMPLETED AS PER ERECTION PROCEDURES AND DRAWINGS.		X	
THE FINAL PRODUCT HAS BEEN INSPECTED AT THE END OF THE JOB AND CONFORMS WITH DRAWINGS AND SPECIFICATIONS.		X	
ANY NON-CONFORMANCES HAVE BEEN CORRECTED AS PER THE APPROVED DISPOSITIONS (write the NC's numbers below this table).		X	
ALL QA REPORTS ARE COMPLETED AS PER THE ITP.	X 		
ALL FORMS HAVE BEEN REVISED AND APPROVED BY THE QA REPRESENTATIVE.		X	
ENGINEERING APPROVAL HAS BEEN OBTAINED WHENEVER REQUIRED.		X	

Comments: Non-conformity: SNC-GMS-04-003



ITP Step # 6300-25 (rotor resistance test) was not completed as it will be completed during commissioning phase.

NCR #: _____

Measurement Instruments	T°	Measured by: _____	dd-mmm-yyyy
Before: _____			
After: _____		ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>27-Sep-2012</u>
		Cust. Repres. _____	